

Gurukiran Reddy Kasireddy

reddykasireddy714@gmail.com | +1(848)-628-4851 | LinkedIn | GitHub | Portfolio Website

Professional Summary

AI Engineer and Machine Learning Engineer with 3+ years of experience building and deploying production-scale AI systems across healthcare, retail, and finance domains. Skilled in LLMs, retrieval-augmented generation (RAG), and agentic AI architectures for developing context-aware applications. Proven ability to design and scale ML pipelines processing 10M+ records/day, achieving up to 50% reduction in data processing time. Hands-on experience in fine-tuning LLMs (PEFT/LoRA), integrating vector databases, and deploying models using MLOps practices on AWS, GCP, and Azure. Strong foundation in NLP, deep learning, and end-to-end ML lifecycle with a focus on scalability, performance, and reliability.

Technical Skills

- **Machine Learning & AI:** LLMs, NLP, Deep Learning, Supervised & Unsupervised Learning, Time Series, Predictive Modeling, Statistical Learning, Model Evaluation, Feature Engineering
- **Generative AI & LLMs:** GPT-4, Gemini, LLaMA-3, Claude, Hugging Face Transformers, RAG, Fine-tuning (PEFT/LoRA), Prompt Engineering, Agentic & Multi-Agent Systems, LLM Evaluation, Guardrails, Hallucination Mitigation
- **Frameworks & Tools:** PyTorch, TensorFlow/Keras, Scikit-learn, LangChain, LangGraph, LlamaIndex, AutoGen, SpaCy, FastAPI, MLflow, Weights & Biases
- **Programming & Data:** Python, SQL, PySpark, Java, R, Apache Kafka, Apache Airflow, ETL Pipelines, Data Engineering, EDA
- **Cloud & MLOps:** AWS (SageMaker, EC2, S3), GCP, Azure, Docker, Kubernetes, CI/CD, Model Deployment, Model Monitoring, Data Drift Detection, Databricks, vLLM, Triton Inference Server
- **Databases & Analytics:** PostgreSQL, MongoDB, Vector DBs (FAISS, Pinecone, Chroma, Weaviate), Snowflake, Power BI, Tableau
- **System Design & APIs:** Distributed Systems, Microservices, REST APIs, Scalable Architecture, Low-Latency Inference, High-Availability Systems

Experience

AI Engineer, UnitedHealth Group – USA

Aug 2025 – Present

- Architected and deployed LLM-powered member support chatbots using GPT-4 and NLP pipelines, automating 70%+ of high-volume healthcare queries and improving resolution efficiency by 40% across digital platforms.
- Engineered hybrid RAG-based clinical retrieval systems integrating structured EHR and unstructured clinical data, enabling real-time query resolution with semantic search and reranking for physicians and members.
- Developed predictive risk stratification models on claims and EHR datasets using XGBoost and Deep Learning, identifying high-risk patients with 85%+ precision and supporting value-based care initiatives.
- Designed and optimized distributed MLOps pipelines (PySpark, SQL, Apache Airflow) processing 10M+ healthcare records/day, reducing data processing latency by 50% and improving ML workflow efficiency.
- Implemented LLM fine-tuning (PEFT/LoRA) with prompt engineering and vector database integration (FAISS, Pinecone) for structured query generation and efficient Top-K retrieval, reducing hallucinations by 35%.
- Deployed and scaled production-grade agentic AI services on AWS (SageMaker, EC2) supporting 10K+ concurrent users, ensuring low-latency inference, high availability, and 99.9% system reliability.

Machine Learning Engineer, CitiusTech Inc – India

May 2021 – Jul 2023

- Architected clinical risk stratification models (XGBoost, Logistic Regression) on EHR and claims data, improving early detection by 22%, reducing hospital readmissions by 18%, and supporting value-based care programs.
- Built population health analytics pipelines processing 5M+ patient records, enabling cohort segmentation, care gap analysis, and actionable insights for clinical decision support teams.
- Designed and implemented FHIR/HL7-based data ingestion pipelines, standardizing clinical data across 10+ healthcare systems and improving interoperability and downstream ML feature quality.
- Engineered end-to-end ML pipelines for patient outcome prediction using temporal and clinical features, improving model AUC by 15% through feature engineering and hyperparameter optimization.
- Implemented A/B testing, real-time model monitoring, and data drift detection systems using MLflow, improving model stability by 20%+ and ensuring reliable production performance across deployments.
- Optimized ML inference pipelines on AWS (SageMaker, EC2), reducing infrastructure costs by 25%, cutting inference latency by 30%, and improving throughput for real-time clinical applications.

Education

M.S. in Computer & Information Systems, New Jersey Institute of Technology (NJIT)

Sep 2023 – May 2025

B.Tech in Computer Science & Engineering, PES University

Aug 2018 – May 2022